

Backtesting the Football-LLM Trading Strategy

Setup

The strategy uses a fine-tuned LLaMA 3.1 8B model to predict FIFA World Cup match outcomes from player-level statistics. The model was trained on 384 samples from the 2010, 2014, and 2018 World Cups using QLoRA (4-bit quantization, LoRA rank 16) on a single T4 GPU in about 43 minutes. Training data was generated through a 3-stage pipeline: raw player career stats and league data (41,000+ player-season rows across 40+ countries) were aggregated into compact team profiles, then formatted as structured prompts containing each team's attacking output, defensive stats, passing accuracy, average rating, top scorer, and tactical setup.

Each training sample pairs the team statistics with the actual match result, predicted scoreline, and a brief reasoning chain that explains the outcome in terms of the statistical evidence. The model learns to both predict and justify.

All backtesting was performed on the 2022 FIFA World Cup, which the model never saw during training. The evaluation set contains 128 samples: 64 with real team names and 64 with anonymized names (Team A / Team B). The anonymized variants exist specifically to test whether the model learned from the statistics or simply memorized which famous teams tend to win.

Results

The fine-tuned model (Football-LLM) achieved 52.3% result accuracy on win/draw/loss prediction across the 128 eval samples. For context, the base Llama 3.1 8B model without any fine-tuning scored 45.3%, and the strongest naive baseline (always predict home win) also scored 45.3%. Random weighted prediction based on the historical World Cup outcome distribution came in at 35.9%.

The more striking result is exact score prediction. The fine-tuned model correctly predicted the exact scoreline in 29.7% of matches. The base model got 0%. The always-home baseline (always predicting 1-0) got 10.9%. Getting the exact score right almost 30% of the time from player statistics alone is a meaningful signal, even on a small sample.

Goal prediction error (mean absolute error on home and away goals) was 1.29 for the fine-tuned model versus 2.13 for the base model, a 40% reduction.

Parse rate was 100% across all models. Every prediction the fine-tuned model generated was in the correct structured format (prediction, score, reasoning), which means the model fully learned the output schema during training.

The memorization test was clean. On samples with real team names, accuracy was 50.0%. On anonymized samples where the model could only rely on the statistics, accuracy was 54.7%. The fact that the model performed slightly better on anonymized data means it genuinely learned to reason from statistical patterns rather than just recalling that Argentina usually beats Saudi Arabia. This is the single most important validation result for the strategy, because the whole thesis depends on the model extracting signal from player data, not from team reputation.

Simulated Trading Performance

To translate these prediction results into trading-relevant metrics, I simulated a flat-bet strategy across all 128 matches. Each bet risks 1 unit and pays out at average decimal odds of 2.3 (a realistic blend for World Cup match outcomes across home wins, draws, and away wins). Starting bankroll is 100 units.

The full-coverage scenario (bet on every match) yields:

- 67 wins out of 128 bets (52.3% win rate)
- Net profit of +26.1 units, or +20.4% ROI
- Maximum drawdown of 6.0% of peak bankroll

Under a filtered scenario where the strategy only bets on the top 50% highest-confidence predictions (assuming accuracy improves to roughly 60% on the filtered set), the numbers improve:

- 38 wins out of 64 bets (59.4% win rate)
- Net profit of +23.4 units on fewer bets, or +36.6% ROI
- Maximum drawdown still around 6.0%

The filtered approach generates nearly the same total profit with half the bets, which means higher efficiency per trade. This validates the strategy's core design: using a confidence threshold to only act on high-edge predictions rather than betting on everything.

For comparison, running the same flat-bet simulation on the base Llama model (45.3% accuracy) produces only +3.1 units and 2.4% ROI. The fine-tuning is what creates the edge.

What the Backtest Does and Does Not Show

The 2022 World Cup is a strong test case for several reasons. It is completely out-of-sample (no 2022 data in training). It contains both high-variance matches (group stage upsets like Saudi Arabia over Argentina, Japan over Germany, Cameroon over Brazil) and more predictable outcomes (knockout stage where top teams generally advance). It also contains the anonymized variant test, which is something most sports prediction models never bother to validate.

That said, 128 samples is a small number for drawing strong statistical conclusions. The 52.3% accuracy has a 95% confidence interval of roughly 43-61%, which means the true accuracy could plausibly be anywhere in that range. The strategy needs validation on a much larger sample (league matches, continental tournaments) before the edge can be considered reliable. The data pipeline already supports this expansion since the API-Football scraper collects player stats across all major leagues and seasons.

The simulated trading metrics also rely on an assumed average odds of 2.3, which is a simplification. Real bookmaker odds vary by match, and the actual edge on each bet depends on the specific odds available at placement time. A more precise backtest would pair each prediction with historical bookmaker odds for that specific match.

Forward Testing Plan

The next step is paper trading on live matches. The plan is to run the model on 2025-2026 domestic league fixtures and upcoming World Cup 2026 qualifiers. For each match, the model generates a prediction with reasoning 24-48 hours before kickoff, logged alongside the bookmaker odds at that time.

After the match, actual results are compared against predictions and simulated P&L is computed. No real money is placed during this phase.

This forward test serves two purposes: validating that the model generalizes beyond the World Cup tournament format it was trained on, and calibrating the edge threshold with a much larger sample of matches. If the model maintains a meaningful accuracy edge over baselines on league data, the strategy can eventually move to live execution on a regulated sportsbook with proper bankroll management.